

Analysis General Exam August 20, 2014

Name: _____

Pledge: _____

Signature: _____

To get credit for a problem, you must show all of your reasoning and calculations.

Pr	Score
1	—
2	—
3	—
4	—
5	—
6	—
7	—
8	—

Problem 1

Suppose $f : [0, 1] \rightarrow \mathbb{C}$ is continuous. Find

$$\lim_{k \rightarrow \infty} \int_0^1 kx^{k-1} f(x) dx$$

Prove your result.

Problem 2

- (a) Show that any open subset of $\mathbb{R}$ is a countable disjoint union of open intervals (a, b) (where $-\infty \leq a < b \leq \infty$).
- (b) Show that the σ -algebra generated by the open subsets of $\mathbb{R}$ is the same as the σ -algebra generated by the intervals $[a, b)$ with $-\infty < a < b < \infty$.

Problem 3

Suppose (X, Σ) is a measurable space, and ν and μ are two measures on the σ -algebra Σ with ν absolutely continuous with respect to μ . Suppose ν is a finite measure. Show that $\nu(E) \rightarrow 0$ as $\mu(E) \rightarrow 0$. (In other words, given $\epsilon > 0$ there exists $\delta > 0$ such that if $E \in \Sigma$ with $\mu(E) < \delta$, then $\nu(E) < \epsilon$.)

[Hint: If not show $\exists \epsilon > 0$ and $E_j \in \Sigma$ such that $\bigcup_{j=1}^{\infty} E_j < \infty$ and $\nu(E_j) \geq \epsilon$. Proceed from there.]

Problem 4

- (a) Use the Monotone Convergence Theorem and $\int_1^t \frac{dx}{x} = \log t$ to show

$$\lim_{n \rightarrow \infty} n \log \left(1 + \frac{t}{n} \right) = t \text{ for } t \geq 0$$

- (b) Show $\lim_{n \rightarrow \infty} \int_0^n \left(1 + \frac{t}{n} \right)^n e^{-2t} dt = 1$.

- (c) Let $\Gamma(x) = \int_0^{\infty} t^{x-1} e^{-t} dt$ for $x > 0$. Show that

$$\begin{aligned} \Gamma(x) &= \lim_{n \rightarrow \infty} \int_0^n \left(1 - \frac{t}{n} \right)^n t^{x-1} dt \\ &= \lim_{n \rightarrow \infty} n^x n! (x(x+1) \cdots (x+n))^{-1} \end{aligned}$$

Problem 5

- (a) State the Mean Value Property for analytic functions, then use the Cauchy Integral Formula to prove it.
- (b) TRUE or FALSE: If u is a harmonic function on a domain in $\mathbb{R}^2$, then u has a harmonic conjugate.
- (c) Find a fractional linear transformation (also called a Möbius transformation) f that takes the first quadrant to the top half of the unit disk and satisfies $f(2) = i$. (You must explain some comprehensible procedure and not simply produce an f out of thin air.) Under your map, what is the image of the vertical ray $\{\operatorname{Re} z = c > 0, \operatorname{Im} z > 0\}$?

Problem 6

Let $f(z)$ be a bounded analytic function in the upper half-plane that extends continuously to the real axis. If $|f(z)| \leq M$ for real z , show that $|f(z)| \leq M$ for all z in the upper half-plane.

[Suggestion: for the top half of an arbitrary disk centered at the origin, consider an appropriate branch of the function $(z + i)^{-\varepsilon} f(z)$ for small enough $\varepsilon > 0$.]

Problem 7

Use the argument principle to determine the number of roots of $p(z) = z^9 + 4z^5 - 3z^4 + 4z + \alpha$ in the right half-plane. The answer may depend on the value of α , which is assumed real.

Problem 8

Let a and b be unequal positive numbers. By integrating an appropriate branch of $\frac{(\log z)^2}{(z+a)(z+b)}$ around a keyhole contour, find

$$\int_0^\infty \frac{\log x}{(x+a)(x+b)} dx$$